

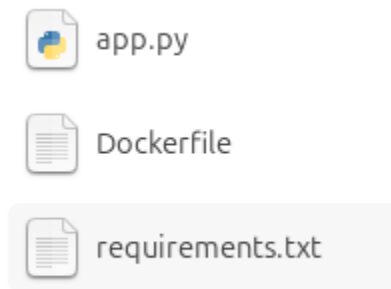
PROGRAM 3: Code a dockerized python flask or [Node.js](#) application

NAME: Ananya Ganapati Hegde

USN: 1RV24MC017

SECTION: A

Directory Structure:



Step 1: Create a Dockerfile as following

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p3$ nano Dockerfile
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p3$ cat Dockerfile
FROM python:latest

WORKDIR app/

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p3$
```

Step 2: Create [app.py](#)

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p3$ nano app.py
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p3$ cat app.py
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return "Hello from dockerized flask application."

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p3$
```

Step 3: Create requirements.txt

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/Dev0ps/p3$ nano requirements.txt
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/Dev0ps/p3$ cat requirements.txt
flask==2.3.3
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/Dev0ps/p3$
```

Step 4: Build the docker image

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p3$ sudo docker build -t program_3 .
[+] Building 4.5s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 198B
=> [internal] load metadata for docker.io/library/python:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e62056bbc28028afd858185d73bf4d6a439cbb058b6800a96d
=> [internal] load build context
=> => transferring context: 484B
=> CACHED [2/5] WORKDIR app/
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:1f2a33b1b0db1cffd03ca8f7e18b30c390d4ca4da98b5624813487382924a9f0
=> => naming to docker.io/library/program_3

1 warning found (use docker --debug to expand):
- WorkdirRelativePath: Relative workdir "app/" can have unexpected results if the base image changes (line 3)
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p3$
```

Step 5: Run the docker image

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p3$ sudo docker run -p 5000:5000 program_3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
```

Step 6: Go to <http://127.0.0.1:5000/> or <http://172.18.0.2:5000/> in the browser

